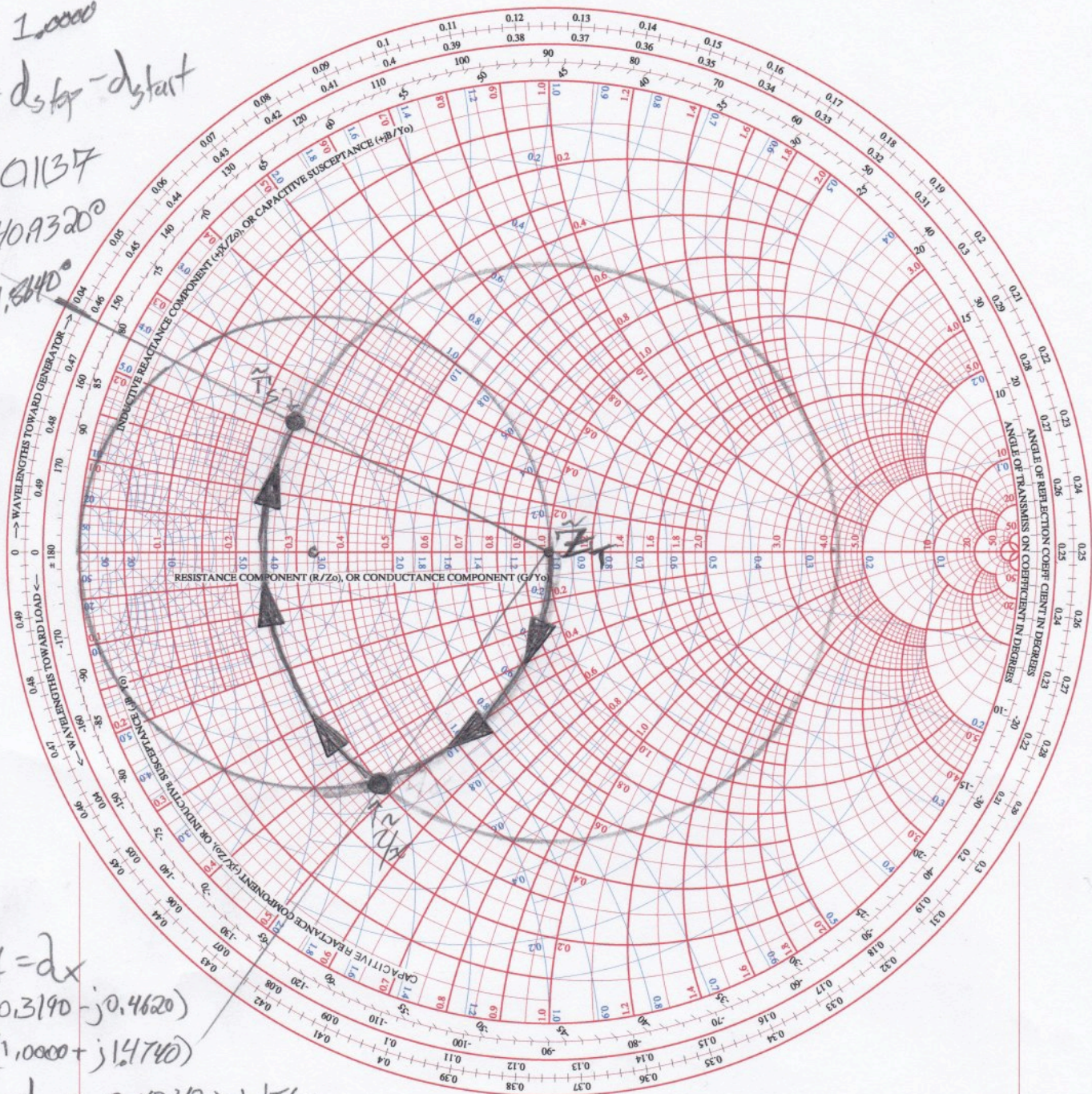


NAME Steven Placcek	TITLE EE 456-Design Project 1	DWG. NO. IMN 1
SMITH CHART FORM ZY-01-N	COLOR BY J. COLVIN, UNIVERSITY OF FLORIDA, 1997	DATE 2/22/25

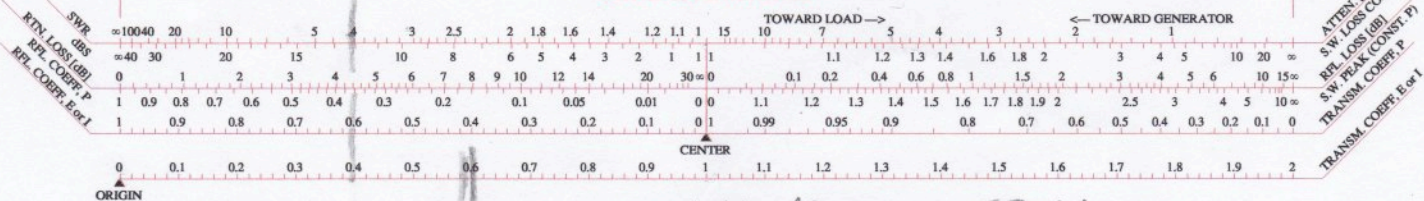
NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

$$\begin{aligned} \frac{Z}{Z_0} &= 1.0000 \\ \frac{Y}{Y_0} &= 1.0000 \\ d_{02} &= d_{stop} - d_{start} \\ d_{02} &= 0.1137 \\ \theta_{02} &= 40.9320^\circ \\ 2\theta_{02} &= 81.8640^\circ \end{aligned}$$



$$\begin{aligned} d_{start} &= d_x \\ Z_x &= (0.3190 - j0.4620) \\ Y_x &= (1.0000 + j1.4174) \\ d_x &= 0.4242 \lambda \text{ WTG} = \end{aligned}$$

RADIALLY SCALED PARAMETERS



$$T_L = (0.5871 + j152.4480)$$

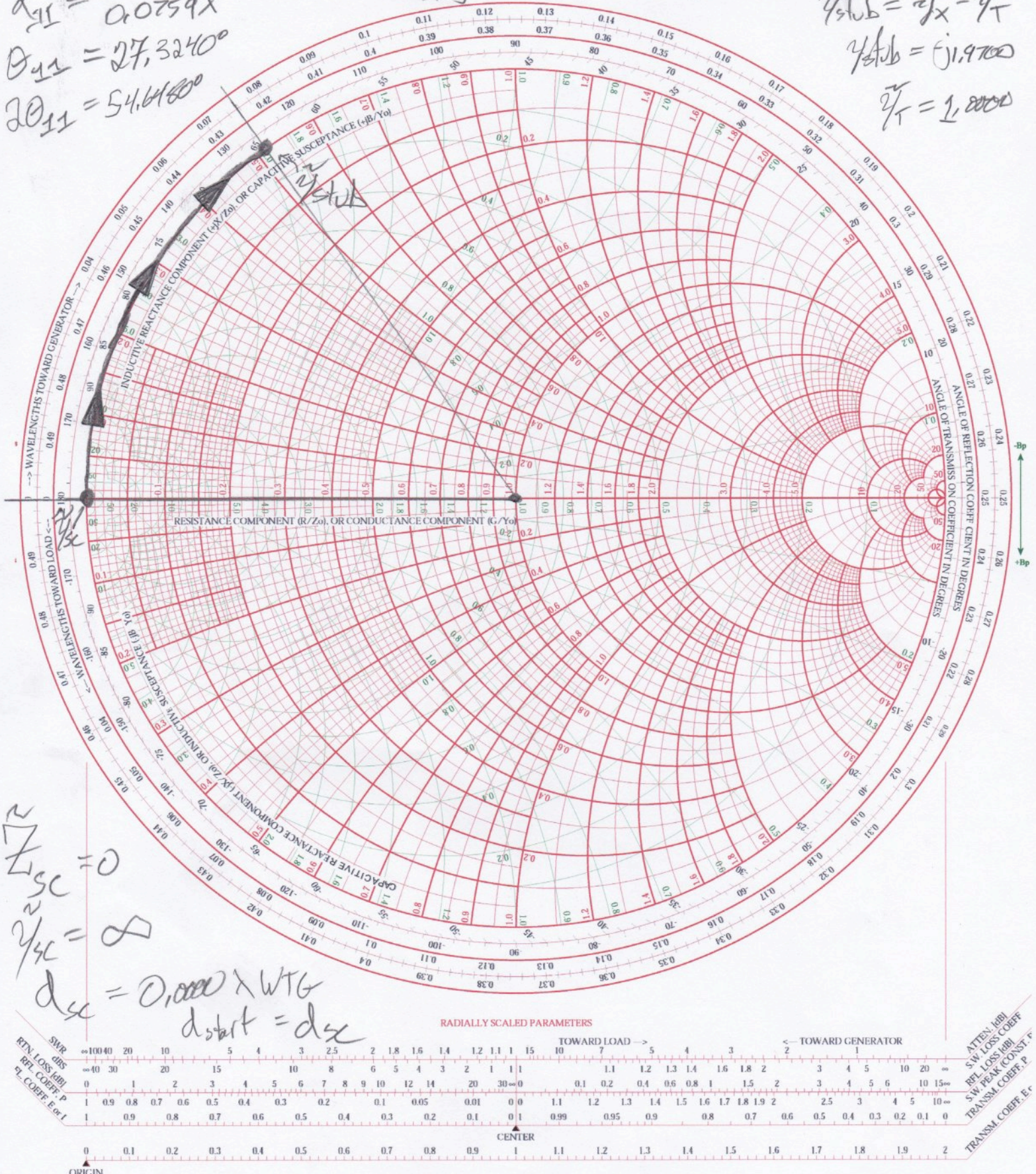
$$VSWR = 0.5871$$

NAME Steven Paezck	TITLE EE 456-Design Project 1	DWG. NO. IMN 2
		DATE 2/22/25

$d_{11} = d_{stop} - d_{start}$
 $d_{11} = 0.0759\lambda$
 $\theta_{11} = 27.3240^\circ$
 $2\theta_{11} = 54.6480^\circ$

$d_{stop} = d_{stub}$
 $d_{stub} = 0.0759 \text{ WTG}$
NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES

$\tilde{Y}_X = 1.000 - j1.9700$
 $\tilde{Y}_{stub} = \tilde{Y}_X - \tilde{Y}_T$
 $\tilde{Y}_{stub} = j1.9700$
 $\tilde{Y}_T = 1.0000$



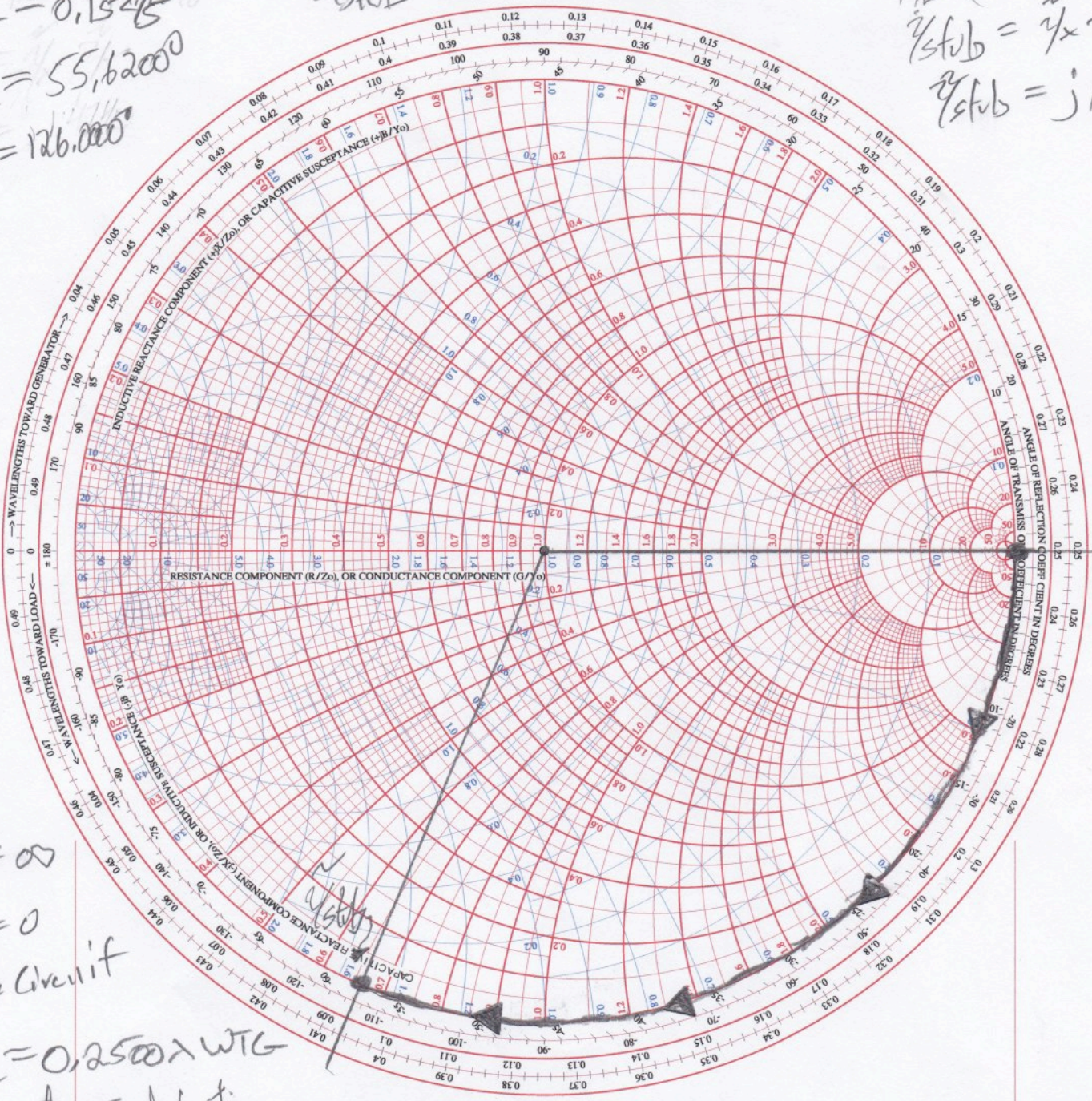
NAME Steven Placzek	TITLE EE 456 - Design Project 1	DWG. NO. OMN 2
SMITH CHART FORM ZY-01-N	COLOR BY J. COLVIN, UNIVERSITY OF FLORIDA, 1997	DATE 2/22/25

$d_{01} = 0.1545$
 $\theta_{01} = 55.6200^\circ$
 $2\theta_{01} = 126.000^\circ$

$d_{stop} = d_{stub} \lambda / 4 = 1.0000$
NORMALIZED IMPEDANCE AND ADMITTANCE COORDINATES
 $d_{stub} = 0.4045 \lambda$ WTG

$\tilde{Y}_x = (1.0000 + j1.4740)$
 $\tilde{Y}_{stub} = \tilde{Y}_x - \tilde{Y}_T$
 $\tilde{Y}_{stub} = j1.4740$

$Z_{oc} = \infty$
 $Y_{oc} = 0$
 Open Circuit
 $d_{oc} = 0.2500 \lambda$ WTG
 $d_{oc} = d_{start}$



$\tilde{Y}_{oc}$

